

DOURI: Forward State-Space Bridges for Offline Goal-Conditioned Reinforcement Learning

Method Draft

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Abstract

DOURI is an offline goal-conditioned reinforcement learning method that learns state-space proposals rather than an explicit environment model. Given a current state and a goal, it predicts an intermediate subgoal, constructs an endpoint-conditioned forward bridge in latent state space, decodes the proposed state transitions into action chunks by inverse dynamics, and trains a short-prefix actor by a critic-weighted proximal objective. The bridge is defined in coordinates centered at the current state and uses a linear-SDE reference process. This reference admits closed-form Gaussian marginals, Doob/SOC controls, and one-step posterior moments. We present the bridge in a compact interval notation and derive the coefficients in the appendix. We also introduce a prefix-calibrated schedule that directly controls the hard-bridge marginal progress visible to the actor’s short executed prefix, and an optional endpoint-preserving path residual that lets the bridge follow non-linear dataset-supported paths without moving either endpoint. All components are optimized jointly under one objective; separate losses are only a notational factorization of the concurrent training procedure.

1 Introduction

Long-horizon offline goal-conditioned reinforcement learning requires agents to compose useful behaviors from fixed datasets. Purely one-step temporal-difference learning can propagate sparse rewards slowly, while autoregressive learned models can accumulate error over long rollouts. Recent action-chunk approaches reduce effective horizon lengths by evaluating or predicting temporally extended action sequences, but long open-loop policies can sacrifice reactivity [5, 6, 7]. Model-based value expansion with action chunks similarly reduces recursive model calls, yet still reports difficulty in contact-rich locomotion domains where model errors accumulate [4].

DOURI follows a complementary route. Instead of learning an environment dynamics model or predicting a long action sequence directly, it proposes a short state path from the current state to a selected subgoal. The path is decoded into actions by inverse dynamics, scored by chunk critics, and distilled into a short-prefix actor. The bridge is therefore a proposal distribution: it should be endpoint-aware, temporally coherent, and dataset-compatible, but it is not assumed to equal the physical transition kernel.

The bridge component is inspired by diffusion bridges. GOUB constructs point-to-point bridges by applying Doob’s h -transform to a generalized Ornstein–Uhlenbeck process and derives exact Gaussian bridge posteriors for training [2]. UniDB interprets such bridges through stochastic optimal control (SOC), with hard Doob bridges recovered as the infinite terminal-penalty limit [3]. DOURI transfers this bridge/SOC viewpoint from image restoration to forward state-space proposal learning. The action side follows the horizon-reduction principle of action chunking: a long critic horizon can accelerate value propagation, while the learned actor executes only a short prefix for closed-loop reactivity [6, 5].

2 Problem Formulation

Let $s_t \in \mathbb{R}^d$ denote the current latent state and s_g the global goal. DOURI first predicts or samples an intermediate subgoal $s_K \in \mathbb{R}^d$, then constructs a forward bridge

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_K, \quad s_0 := s_t.$$

We work in displacement coordinates

$$r_i := s_i - s_0, \quad \delta := s_K - s_0. \quad (1)$$

For a hard bridge, $r_0 = 0$ and $r_K = \delta$. The bridge horizon is K . The actor may use only a prefix of length $h_a \leq K$, and the full chunk critic may use a longer horizon H . This yields the high-level pipeline

$$(s_t, s_g) \xrightarrow{\text{subgoal}} \hat{s}_K \xrightarrow{\text{state bridge}} \hat{s}_{0:K} \xrightarrow{\text{inverse dynamics}} \{\hat{a}_{0:h_a-1}^{(m)}\}_{m=1}^M \xrightarrow{Q\text{-Boltzmann}} \pi_{\text{invdyn}}^Q \xrightarrow{\text{SPI}} \mu_\phi.$$

All modules are trained concurrently. The separate components below define the terms of a single joint objective.

3 Forward Linear-SDE Bridge

3.1 Passive reference dynamics

DOURI defines a passive bridge-time reference process in displacement coordinates:

$$dr_t = \theta_t r_t dt + g_t dW_t, \quad g_t^2 = 2\lambda^2 \theta_t. \quad (2)$$

For a discrete interval $i \rightarrow i+1$, let $\theta_i > 0$ be the integrated rate. The exact discretization is

$$r_{i+1} = A_i r_i + \eta_i, \quad A_i := e^{\theta_i}, \quad \eta_i \sim \mathcal{N}(0, q_i^2 I), \quad (3)$$

where

$$q_i^2 = \lambda^2 (e^{2\theta_i} - 1). \quad (4)$$

The exponential factor is the exact homogeneous flow of $\dot{r}_t = \theta_t r_t$. We keep $g_i^2 = 2\lambda^2 \theta_i$ and q_i^2 distinct: q_i^2 is used in exact discrete posterior moments, while g_i^2 is retained as the continuous-time reference coefficient and reverse-sampling scale.

3.2 Interval notation

For $0 \leq i \leq j \leq K$, define

$$\Theta_{i:j} := \sum_{\ell=i}^{j-1} \theta_\ell, \quad F_{i:j} := e^{\Theta_{i:j}}, \quad S_{i:j} := \lambda^2 (F_{i:j}^2 - 1), \quad (5)$$

with $\Theta_{i:i} = 0$, $F_{i:i} = 1$, and $S_{i:i} = 0$. These quantities are respectively the accumulated bridge-time rate, deterministic interval flow, and accumulated interval variance.

Proposition 1 (Passive interval transition). *Under the passive reference, for all $0 \leq i \leq j \leq K$,*

$$r_j \mid r_i \sim \mathcal{N}(F_{i:j} r_i, S_{i:j} I). \quad (6)$$

In particular, since $r_0 = 0$, $r_i \sim \mathcal{N}(0, S_{0:i} I)$.

The proof is given in Appendix A. The same proof also gives the decomposition

$$S_{0:K} = F_{i:K}^2 S_{0:i} + S_{i:K}. \quad (7)$$

3.3 Soft terminal bridge

DOURI allows a finite terminal penalty

$$\frac{\gamma}{2} \|r_K - \delta\|_2^2, \quad \varepsilon_\gamma := \gamma^{-1}. \quad (8)$$

The hard Doob bridge corresponds to $\varepsilon_\gamma = 0$. Equivalently, finite γ is a noisy terminal observation model

$$\delta = r_K + \zeta, \quad \zeta \sim \mathcal{N}(0, \varepsilon_\gamma I). \quad (9)$$

Proposition 2 (Bridge marginal). *The bridge marginal at step i is*

$$r_i \mid \delta \sim \mathcal{N}(\beta_i^\gamma \delta, B_i^\gamma I), \quad (10)$$

where

$$\beta_i^\gamma = \frac{S_{0:i} F_{i:K}}{S_{0:K} + \varepsilon_\gamma}, \quad B_i^\gamma = \frac{S_{0:i}(S_{i:K} + \varepsilon_\gamma)}{S_{0:K} + \varepsilon_\gamma}. \quad (11)$$

Equivalently, in state coordinates,

$$q_i^\gamma(s_i \mid s_0, s_K) = \mathcal{N}((1 - \beta_i^\gamma)s_0 + \beta_i^\gamma s_K, B_i^\gamma I). \quad (12)$$

For the hard bridge, $\beta_0 = 0$, $B_0 = 0$, $\beta_K = 1$, and $B_K = 0$.

Proposition 3 (One-step bridge posterior). *The exact one-step bridge transition is Gaussian:*

$$r_{i+1} \mid r_i, \delta \sim \mathcal{N}(\bar{r}_{i+1}^{\star, \gamma}, C_i^\gamma I). \quad (13)$$

Let

$$\kappa_i^\gamma := \frac{q_i^2 F_{i+1:K}}{S_{i:K} + \varepsilon_\gamma}. \quad (14)$$

Then

$$\bar{r}_{i+1}^{\star, \gamma} = A_i r_i + \kappa_i^\gamma (\delta - F_{i:K} r_i), \quad C_i^\gamma = q_i^2 \frac{S_{i+1:K} + \varepsilon_\gamma}{S_{i:K} + \varepsilon_\gamma}. \quad (15)$$

Thus the state-coordinate teacher mean is

$$\mu_i^{\star, \gamma} = s_0 + e^{\theta_i} (s_i - s_0) + \kappa_i^\gamma [(s_K - s_0) - F_{i:K} (s_i - s_0)]. \quad (16)$$

For a hard bridge, $C_{K-1} = 0$ and $\mu_{K-1}^\star = s_K$.

The derivations of Propositions 2 and 3 are provided in Appendices B and C.

3.4 Doob/SOC control

The soft h -function is

$$h_i^\gamma(r; \delta) := \mathbb{E}_R \left[\exp \left(-\frac{\gamma}{2} \|r_K - \delta\|_2^2 \right) \mid r_i = r \right]. \quad (17)$$

Since $r_K \mid r_i = r \sim \mathcal{N}(F_{i:K} r, S_{i:K} I)$,

$$h_i^\gamma(r; \delta) \propto \exp \left(-\frac{1}{2} \frac{\|\delta - F_{i:K} r\|_2^2}{S_{i:K} + \varepsilon_\gamma} \right), \quad (18)$$

and therefore

$$\nabla_r \log h_i^\gamma(r; \delta) = \frac{F_{i:K}}{S_{i:K} + \varepsilon_\gamma} (\delta - F_{i:K} r). \quad (19)$$

The corresponding controlled drift contribution is

$$u_i^{h, \gamma}(r_i) = g_i^2 \frac{F_{i:K}}{S_{i:K} + \varepsilon_\gamma} (\delta - F_{i:K} r_i). \quad (20)$$

The current transition model uses the exact posterior moments in Proposition 3 directly; Eq. (20) is included to expose the equivalent Doob/SOC form of the same bridge. Appendix D gives the SOC derivation and its connection to Doob's h -transform.

4 Gauge-Translated Displacement Bridge

The forward bridge may be written directly in state coordinates, but DOURI uses a translated displacement chart anchored at the current state. This is analogous to applying a gauge transformation: the bridge is developed in local coordinates r , while all environment-facing outputs are mapped back to state space by adding the anchor state.

For an offline segment

$$(s_t^{\mathcal{D}}, s_{t+1}^{\mathcal{D}}, \dots, s_{t+K}^{\mathcal{D}}),$$

define the anchor and displacement trajectory

$$s_0 := s_t^{\mathcal{D}}, \quad r_i^{\mathcal{D}} := s_{t+i}^{\mathcal{D}} - s_t^{\mathcal{D}}, \quad r_0^{\mathcal{D}} = 0, \quad \delta^{\mathcal{D}} := r_K^{\mathcal{D}}. \quad (21)$$

At inference time, for an arbitrary current state s_t , the inverse chart is

$$\mathcal{T}_{s_t}(r) := s_t + r. \quad (22)$$

Thus the bridge never needs to output an absolute state internally. It outputs a displacement path

$$\hat{r}_0, \hat{r}_1, \dots, \hat{r}_K, \quad \hat{r}_0 = 0,$$

and the corresponding proposed state path is reconstructed as

$$\hat{s}_{t+i} = s_t + \hat{r}_i, \quad i = 0, \dots, K. \quad (23)$$

In particular, the terminal state proposal is

$$\hat{s}_{t+K}^{\text{out}} = s_t + \hat{r}_K. \quad (24)$$

This coordinate choice separates the bridge geometry from absolute state coordinates. The analytic bridge lives in r -space, while the critic, inverse dynamics, and value functions continue to receive ordinary state-space inputs through Eq. (23).

4.1 Terminal displacement proposal

The subgoal module is written as a proposal over terminal displacements rather than absolute future states:

$$\hat{\delta} \sim q_{\chi}^r(\cdot \mid s_t, s_g). \quad (25)$$

The corresponding state-space subgoal is

$$\hat{s}_{t+K} = s_t + \hat{\delta}. \quad (26)$$

For a deterministic subgoal estimator, this becomes

$$\hat{\delta} = G_{\chi}^r(s_t, s_g), \quad \hat{s}_{t+K} = s_t + G_{\chi}^r(s_t, s_g).$$

For a stochastic Gaussian estimator, for example,

$$q_{\chi}^r(\delta \mid s_t, s_g) = \mathcal{N}(\mu_{\chi}^r(s_t, s_g), \text{diag}((\sigma_{\chi}^r)^2(s_t, s_g))),$$

with

$$\hat{\delta} = \mu_{\chi}^r(s_t, s_g) + \sigma_{\chi}^r(s_t, s_g) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The terminal proposal loss is correspondingly written in displacement coordinates:

$$\mathcal{L}_{\text{sg}}^r = \mathbb{E}_{(s_t, s_{t+K}^{\mathcal{D}}, s_g) \sim \mathcal{D}} \left[\left\| \hat{\delta} - (s_{t+K}^{\mathcal{D}} - s_t) \right\|_2^2 - \alpha_{\text{sg}} V_{\xi}(s_t + \hat{\delta}, s_g) \right]. \quad (27)$$

The regression term keeps the terminal displacement dataset-supported, while the value term evaluates the reconstructed terminal state $s_t + \hat{\delta}$.

4.2 Bridge evolution in displacement coordinates

Conditioned on a terminal displacement δ , the analytic bridge marginal from Section 3 is

$$r_i \mid \delta \sim \mathcal{N}(\beta_i^\gamma \delta, B_i^\gamma I). \quad (28)$$

Equivalently, after applying the translation map,

$$s_{t+i} \mid s_t, \delta \sim \mathcal{N}(s_t + \beta_i^\gamma \delta, B_i^\gamma I). \quad (29)$$

The one-step bridge teacher is also kept in displacement coordinates:

$$\bar{r}_{i+1}^{\star, \gamma} = A_i r_i + \kappa_i^\gamma (\delta - F_{i:K} r_i), \quad C_i^\gamma = q_i^2 \frac{S_{i+1:K} + \epsilon_\gamma}{S_{i:K} + \epsilon_\gamma}. \quad (30)$$

The state-coordinate teacher is obtained only after reconstruction:

$$\mu_i^{\star, \gamma, s} = s_t + \bar{r}_{i+1}^{\star, \gamma}. \quad (31)$$

For a hard bridge, the final step satisfies

$$C_{K-1} = 0, \quad \bar{r}_K^\star = \delta, \quad s_t + \bar{r}_K^\star = s_t + \delta.$$

Thus the endpoint is pinned in r -space and is mapped back to the absolute subgoal by Eq. (22).

4.3 Parameterized displacement transition

The learned bridge transition predicts the next displacement:

$$\hat{r}_{i+1} = M_{\psi, i}^r(\hat{r}_i, s_t, \delta), \quad \hat{r}_0 = 0. \quad (32)$$

The corresponding state transition used by downstream modules is

$$\hat{s}_{t+i+1} = s_t + M_{\psi, i}^r(\hat{r}_i, s_t, \delta). \quad (33)$$

A convenient exact-residual parameterization uses the analytic bridge posterior mean as a base transition:

$$M_{\psi, i}^r(r_i, s_t, \delta) = \bar{r}_{i+1}^{\star, \gamma}(r_i, \delta) + \alpha_{\text{res}} \sqrt{C_i^\gamma} R_\psi(r_i, s_t, \delta, i). \quad (34)$$

The factor $\sqrt{C_i^\gamma}$ makes the residual vanish as the bridge becomes deterministic. In particular, for a hard bridge at the final step, $C_{K-1} = 0$, so

$$M_{\psi, K-1}^r(r_{K-1}, s_t, \delta) = \delta, \quad \hat{s}_{t+K} = s_t + \delta,$$

independently of the residual network.

An SDE-style score/residual parameterization can also be written entirely in r -space:

$$M_{\psi, i}^r(r_i, s_t, \delta) = r_i + \theta_i r_i + g_i^2 \frac{F_{i:K}}{S_{i:K} + \epsilon_\gamma} (\delta - F_{i:K} r_i) - \rho_i \epsilon_\theta(r_i, s_t, \delta, i), \quad (35)$$

where ρ_i is the residual scaling coefficient. In the implementation, Eq. (34) and Eq. (35) are two choices for the same displacement transition interface. All state outputs remain $s_t + M^r(\cdot)$.

4.4 Bridge and trajectory losses in the translated chart

The analytic bridge loss matches the predicted next displacement to the exact one-step bridge teacher:

$$\mathcal{L}_{\text{br}}^r = \mathbb{E}_{i \sim \text{Unif}\{0, \dots, K-1\}} \left[\frac{1}{2g_i^2} \left\| \bar{r}_{i+1}^{\star, \gamma}(r_i, \delta) - M_{\psi, i}^r(r_i, s_t, \delta) \right\|_1 \right]. \quad (36)$$

$\delta \sim \delta^{\mathcal{D}}$
 $r_i \sim q_i^{\gamma}(\cdot | \delta)$

This term is best interpreted as a weak bridge-consistency regularizer when a data-dependent residual is used. A large weight on $\mathcal{L}_{\text{br}}^r$ would collapse the residual toward the analytic bridge and remove the path-shape correction learned from data.

The main path-supervision loss is written on dataset displacements:

$$\mathcal{L}_{\text{path}}^r = \mathbb{E}_{\substack{(s_t^{\mathcal{D}}, \dots, s_{t+K}^{\mathcal{D}}) \sim \mathcal{D} \\ i \sim \text{Unif}\{0, \dots, K-1\}}} \left[\left\| r_{i+1}^{\mathcal{D}} - M_{\psi, i}^r(r_i^{\mathcal{D}}, s_t^{\mathcal{D}}, \delta^{\mathcal{D}}) \right\|_1 \right]. \quad (37)$$

Equivalently, in reconstructed state coordinates,

$$\mathcal{L}_{\text{path}}^r = \mathbb{E} \left[\left\| s_{t+i+1}^{\mathcal{D}} - (s_t^{\mathcal{D}} + M_{\psi, i}^r(r_i^{\mathcal{D}}, s_t^{\mathcal{D}}, \delta^{\mathcal{D}})) \right\|_1 \right]. \quad (38)$$

The two forms are identical because the same anchor $s_t^{\mathcal{D}}$ is added to both sides.

For rollout supervision, recursively define

$$\hat{r}_0 = 0, \quad \hat{r}_{i+1} = M_{\psi, i}^r(\hat{r}_i, s_t^{\mathcal{D}}, \delta^{\mathcal{D}}). \quad (39)$$

The rollout loss is

$$\mathcal{L}_{\text{roll}}^r = \frac{1}{H_{\text{roll}}} \sum_{i=1}^{H_{\text{roll}}} \left\| \hat{r}_i - r_i^{\mathcal{D}} \right\|_1. \quad (40)$$

Equivalently,

$$\mathcal{L}_{\text{roll}}^r = \frac{1}{H_{\text{roll}}} \sum_{i=1}^{H_{\text{roll}}} \left\| (s_t^{\mathcal{D}} + \hat{r}_i) - s_{t+i}^{\mathcal{D}} \right\|_1.$$

A small residual regularizer is used for the exact-residual model:

$$\mathcal{L}_{\text{res}} = \mathbb{E} \left[\left\| R_{\psi}(r_i, s_t, \delta, i) \right\|_2^2 \right]. \quad (41)$$

4.5 Interface to inverse dynamics and actor extraction

After the displacement path is reconstructed into state space by Eq. (23), inverse dynamics decodes a short action prefix:

$$\hat{a}_i = I_{\omega}(s_t + \hat{r}_i, s_t + \hat{r}_{i+1}), \quad i = 0, \dots, h_a - 1. \quad (42)$$

The inverse dynamics loss remains an ordinary state-action supervised loss:

$$\mathcal{L}_{\text{idm}} = \mathbb{E} \left[\left\| I_{\omega}(s_t, s_{t+1}) - a_t \right\|_2^2 \right]. \quad (43)$$

Thus the bridge is gauge-translated internally, but the action decoder and critic operate on reconstructed physical states.

Given M sampled displacement paths

$$\{\hat{r}_{0:K}^{(m)}\}_{m=1}^M,$$

DOURI reconstructs

$$\hat{s}_{t+i}^{(m)} = s_t + \hat{r}_i^{(m)}$$

and decodes action-prefix candidates

$$\hat{a}_i^{(m)} = I_{\omega}(\hat{s}_{t+i}^{(m)}, \hat{s}_{t+i+1}^{(m)}).$$

These candidates are scored by the partial chunk critic and used in the Q-Boltzmann/SPI actor extraction step described in Section 6.3.

4.6 Training objective

With the translated displacement chart, the bridge-side objective is

$$\mathcal{L}_{\text{disp}} = w_{\text{path}}\mathcal{L}_{\text{path}}^r + w_{\text{roll}}\mathcal{L}_{\text{roll}}^r + w_{\text{br}}\mathcal{L}_{\text{br}}^r + w_{\text{res}}\mathcal{L}_{\text{res}} + w_{\text{sg}}\mathcal{L}_{\text{sg}}^r. \quad (44)$$

The complete joint objective is

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{disp}} + w_{\text{idm}}\mathcal{L}_{\text{idm}} + w_{Q^H}\mathcal{L}_{Q^H} + w_{Q^P}\mathcal{L}_{Q^P} + w_V\mathcal{L}_V + w_{\pi}\mathcal{L}_{\text{SPI}}. \quad (45)$$

Here, $\mathcal{L}_{Q^H}$, $\mathcal{L}_{Q^P}$, $\mathcal{L}_V$, and $\mathcal{L}_{\text{SPI}}$ are unchanged from the critic and actor losses, except that bridge-generated candidates enter those losses only after reconstruction through $s_t + \hat{r}_i$.

The important structural point is that every bridge loss is computed in the translated chart:

$$r_i = s_{t+i} - s_t,$$

whereas every downstream control module receives the reconstructed state:

$$s_{t+i} = s_t + r_i.$$

This keeps the bridge endpoint-conditioned and translation-centered, while preserving the standard state-space interface required by inverse dynamics, critics, and actors.

5 Dynamics Model and Forward-Bridge Planner Variants

5.1 Exact-residual dynamics model

The implemented learned transition model uses the analytic one-step bridge posterior mean as the base transition:

$$\mu_{\psi,i}^{\text{res}} = \mu_i^{\star,\gamma} + \alpha_{\text{res}} \sqrt{C_i^\gamma} r_\psi(s_i, s_0, s_K, i). \quad (46)$$

Since $C_{K-1} = 0$ for the hard bridge, the residual vanishes at the terminal step and the endpoint remains pinned regardless of the residual network. This model is trained primarily by real trajectory supervision. For an offline segment $(s_0^\mathcal{D}, \dots, s_K^\mathcal{D})$,

$$\mathcal{L}_{\text{path}} = \mathbb{E}_i \left[\|s_{i+1}^\mathcal{D} - \mu_i^{\text{model}}(s_i^\mathcal{D}, s_0^\mathcal{D}, s_K^\mathcal{D})\|_1 \right], \quad (47)$$

where μ_i^{model} is Eq. (46). A short rollout loss recursively applies the same model mean from $s_0^\mathcal{D}$:

$$\mathcal{L}_{\text{roll}} = \frac{1}{H_{\text{roll}}} \sum_{i=1}^{H_{\text{roll}}} \|\hat{s}_i - s_i^\mathcal{D}\|_1. \quad (48)$$

The exact-residual model also uses a small residual regularizer $\mathbb{E}\|r_\psi\|_2^2$. A strong bridge-matching term is avoided because it would collapse the residual toward zero and remove the data-dependent correction.

5.2 Closed-form forward-bridge planner

DOURI also supports a path-supervised planner that uses the closed-form bridge marginal mean directly:

$$\bar{s}_i = (1 - \beta_i)s_0 + \beta_i s_K, \quad i = 0, \dots, K, \quad (49)$$

with $\beta_i = \beta_i^\gamma$ from Proposition 2 or the corresponding hard-bridge coefficient when $\varepsilon_\gamma = 0$. This planner does not learn a transition mean; it is trained by path supervision against dataset trajectory segments.

The endpoint-preserving residual planner augments the closed-form path by

$$\hat{s}_i = \bar{s}_i + w_i R_\psi(s_0, s_K, i), \quad w_i := \frac{i(K-i)}{K^2}. \quad (50)$$

Because $w_0 = w_K = 0$, the start and subgoal endpoints are preserved exactly. This residual is a path-shape correction, not an environment dynamics model.

6 Subgoal, Inverse Dynamics, Critic, and Actor

6.1 Value-guided subgoal objective

The subgoal module defines a conditional proposal distribution over future states,

$$q_\chi(s_{t+K} \mid s_t, s_g).$$

This notation covers both deterministic and stochastic subgoal estimators. In the deterministic case, q_χ is a Dirac distribution

$$q_\chi(\cdot \mid s_t, s_g) = \delta_{G_\chi(s_t, s_g)}(\cdot),$$

so that $\hat{s}_{t+K} = G_\chi(s_t, s_g)$. In the stochastic case, for example, q_χ may be a diagonal Gaussian,

$$q_\chi(s_{t+K} \mid s_t, s_g) = \mathcal{N}(\mu_\chi(s_t, s_g), \text{diag}(\sigma_\chi^2(s_t, s_g))),$$

with samples generated by the reparameterization

$$\hat{s}_{t+K} = \mu_\chi(s_t, s_g) + \sigma_\chi(s_t, s_g) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The subgoal objective is

$$\mathcal{L}_{\text{sg}} = \mathbb{E}_{\substack{(s_t, s_{t+K}^{\mathcal{D}}) \sim \mathcal{D} \\ \hat{s}_{t+K} \sim q_\chi(\cdot \mid s_t, s_g)}} \left[\|\hat{s}_{t+K} - s_{t+K}^{\mathcal{D}}\|_2^2 - \alpha_{\text{sg}} V_\xi(\hat{s}_{t+K}, s_g) \right]. \quad (51)$$

The first term keeps sampled subgoals close to dataset-supported future states, while the value term biases the proposal distribution toward subgoals that are useful for reaching the final goal. For deterministic subgoal prediction, the expectation over q_χ collapses to the single prediction $G_\chi(s_t, s_g)$. For stochastic subgoal prediction, the expectation is estimated with one or more reparameterized samples from $q_\chi(\cdot \mid s_t, s_g)$. This objective avoids advantage-weighted likelihood ratios and applies the same value-guided regression principle to both deterministic and stochastic subgoal estimators.

6.2 Inverse dynamics and Q-Boltzmann proposals

Given a proposed path $\hat{s}_{0:K}^{(m)}$, inverse dynamics decodes a candidate action prefix:

$$\hat{a}_i^{(m)} = I_\omega(\hat{s}_i^{(m)}, \hat{s}_{i+1}^{(m)}), \quad i = 0, \dots, h_a - 1. \quad (52)$$

The inverse dynamics loss is

$$\mathcal{L}_{\text{idm}} = \mathbb{E} \|I_\omega(s_t, s_{t+1}) - a_t\|_2^2. \quad (53)$$

Let q_m be the partial-chunk critic score of candidate m . DOURI defines an empirical Q-Boltzmann distribution over inverse-dynamics proposals:

$$\rho_m = \frac{\exp(\beta_\pi q_m)}{\sum_{j=1}^M \exp(\beta_\pi q_j)}, \quad \pi_{\text{invdyn}}^Q(a \mid s_t, s_g) = \sum_{m=1}^M \rho_m \delta_{\hat{a}_{0:h_a-1}^{(m)}}(a). \quad (54)$$

6.3 SPI-style actor extraction

The bridge planner and inverse dynamics model produce M candidate short action chunks

$$\left\{ \hat{a}_{0:h_a-1}^{(m)} \right\}_{m=1}^M, \quad \hat{a}_{0:h_a-1}^{(m)} \in \mathbb{R}^{h_a d_a}.$$

Each candidate is scored by the normalized partial-chunk critic

$$q_m := \bar{Q}_\varphi^{h_a} \left(s_t, \hat{s}_K, \hat{a}_{0:h_a-1}^{(m)} \right),$$

where $\hat{s}_K$ is the bridge endpoint predicted or sampled by the subgoal module. We define the Q-Boltzmann weights

$$\rho_m = \frac{\exp(\beta_{\text{sbi}} \text{sg}(q_m))}{\sum_{j=1}^M \exp(\beta_{\text{sbi}} \text{sg}(q_j))}.$$

These weights induce an empirical inverse-dynamics proposal distribution

$$\pi_{\text{invdyn}}^Q(\cdot \mid s_t, \hat{s}_K) := \sum_{m=1}^M \text{sg}(\rho_m) \delta_{\text{sg}(\hat{a}_{0:h_a-1}^{(m)})}(\cdot).$$

The stop-gradient operators indicate that the actor update does not backpropagate through proposal generation or proposal scoring.

We use the term *single-step proximal policy improvement* (SPI) [1] to denote one proximal improvement step under a critic-induced energy. For a policy distribution $\pi(\cdot \mid s_t, \hat{s}_K)$, define the local energy

$$\mathcal{E}_{s_t, \hat{s}_K}(\pi) := -\mathbb{E}_{a \sim \pi} \left[\bar{Q}_\varphi^{h_a}(s_t, \hat{s}_K, a) \right].$$

A W_2 -SPI update from the reference proposal π_{invdyn}^Q is

$$\pi^+ \in \arg \min_{\pi} \left\{ \mathcal{E}_{s_t, \hat{s}_K}(\pi) + \frac{1}{2\tau} W_2^2 \left(\pi, \pi_{\text{invdyn}}^Q \right) \right\}.$$

Thus the critic term improves the actor, while the Wasserstein proximal term keeps the actor close to the bridge-induced, critic-weighted proposal distribution. This is a single proximal step; DOURI does not perform iterative re-centering or multi-step refinement in the actor update.

In implementation, the actor is deterministic:

$$\pi_\phi(\cdot \mid s_t, \hat{s}_K) = \delta_{\mu_\phi(s_t, \hat{s}_K)}.$$

Restricting the W_2 -SPI objective to this Dirac policy family gives

$$\mu_\phi \in \arg \min_{\mu} \left\{ -\bar{Q}_\varphi^{h_a}(s_t, \hat{s}_K, \mu) + \frac{1}{2\tau} W_2^2 \left(\delta_\mu, \pi_{\text{invdyn}}^Q \right) \right\}.$$

Since the source distribution is a single Dirac mass, the only admissible coupling sends the full mass from μ to the atoms of π_{invdyn}^Q according to their weights ρ_m . Therefore

$$W_2^2 \left(\delta_\mu, \pi_{\text{invdyn}}^Q \right) = \sum_{m=1}^M \text{sg}(\rho_m) \left\| \mu - \text{sg}(\hat{a}_{0:h_a-1}^{(m)}) \right\|_2^2.$$

The actor objective becomes

$$\mathcal{L}_{\text{SPI}} = \mathbb{E} \left[-\bar{Q}_\varphi^{h_a}(s_t, \hat{s}_K, \mu_\phi(s_t, \hat{s}_K)) + \frac{1}{2\tau} \sum_{m=1}^M \text{sg}(\rho_m) \left\| \mu_\phi(s_t, \hat{s}_K) - \text{sg}(\hat{a}_{0:h_a-1}^{(m)}) \right\|_2^2 \right]. \quad (55)$$

The temperature β_{sbi} controls how sharply the proposal distribution concentrates on high-value inverse-dynamics chunks, while τ controls the strength of the Wasserstein proximal anchor. Smaller τ yields a more conservative actor close to bridge-induced proposals; larger τ gives stronger critic-driven improvement.

7 Joint Objective

DOURI optimizes all modules concurrently. A representative joint objective is

$$\begin{aligned} \mathcal{L}_{\text{joint}} = & w_{\text{dyn}}\mathcal{L}_{\text{dyn}} + w_{\text{path}}\mathcal{L}_{\text{path}} + w_{\text{roll}}\mathcal{L}_{\text{roll}} + w_{\text{sg}}\mathcal{L}_{\text{sg}} + w_{\text{idm}}\mathcal{L}_{\text{idm}} \\ & + w_H\mathcal{L}_{QH} + w_P\mathcal{L}_{QP} + w_V\mathcal{L}_V + w_\pi\mathcal{L}_{\text{SPI}}. \end{aligned} \quad (56)$$

For the exact-residual transition model, $\mathcal{L}_{\text{dyn}}$ consists of residual regularization and, optionally, a weak bridge-matching diagnostic term. For the closed-form forward-bridge planners, there is no separate learned transition-mean loss; the planner is trained through path supervision and the subgoal and inverse-dynamics losses. The critic losses are detailed in Appendix G.

8 Discussion

The bridge theorem is schedule-agnostic; any positive θ_i sequence yields closed-form Gaussian coefficients. The schedule is therefore an inductive bias. The prefix-calibrated schedule aligns bridge progress with the actor-visible prefix, avoiding the late-pinning behavior of inherited diffusion schedules when only early proposal steps are used for action extraction. The endpoint-preserving path residual addresses a complementary issue: in structured environments such as mazes, the appropriate endpoint path may be non-Euclidean in raw state space. Together, prefix calibration and endpoint-preserving residual paths make the bridge a more appropriate state proposal for short-prefix, critic-filtered offline control.

A Passive Linear-SDE Derivations

A.1 Exact one-step variance

Consider the scalar-coordinate SDE over one interval with constant integrated rate θ_i :

$$dr_t = \theta_t r_t dt + g_t dW_t, \quad g_t^2 = 2\lambda^2 \theta_t. \quad (57)$$

For an interval normalized so that $\int_i^{i+1} \theta_t dt = \theta_i$, the solution has deterministic flow $A_i = e^{\theta_i}$. The noise variance is

$$\int_i^{i+1} \exp\left(2 \int_u^{i+1} \theta_v dv\right) g_u^2 du. \quad (58)$$

Using $g_u^2 = 2\lambda^2 \theta_u$, this becomes

$$\lambda^2 \left(e^{2\theta_i} - 1\right). \quad (59)$$

Thus $q_i^2 = \lambda^2(e^{2\theta_i} - 1)$.

A.2 Proof of Proposition 1

The one-step transition is

$$r_{i+1} = A_i r_i + \eta_i, \quad \eta_i \sim \mathcal{N}(0, q_i^2 I). \quad (60)$$

Iterating from i to j ,

$$r_j = F_{i:j} r_i + \sum_{\ell=i}^{j-1} F_{\ell+1:j} \eta_\ell. \quad (61)$$

The conditional variance is

$$\sum_{\ell=i}^{j-1} q_\ell^2 F_{\ell+1:j}^2. \quad (62)$$

Substituting $q_\ell^2 = \lambda^2(e^{2\theta_\ell} - 1)$ gives

$$\sum_{\ell=i}^{j-1} \lambda^2(e^{2\theta_\ell} - 1) e^{2\sum_{m=\ell+1}^{j-1} \theta_m} = \lambda^2 \sum_{\ell=i}^{j-1} \left(e^{2\sum_{m=\ell}^{j-1} \theta_m} - e^{2\sum_{m=\ell+1}^{j-1} \theta_m} \right) \quad (63)$$

$$= \lambda^2(e^{2\Theta_{i:j}} - 1) = S_{i:j}. \quad (64)$$

This proves the transition law.

B Bridge Marginal Derivation

Under the passive reference,

$$r_i \sim \mathcal{N}(0, S_{0:i}I), \quad r_K \mid r_i \sim \mathcal{N}(F_{i:K}r_i, S_{i:K}I). \quad (65)$$

With the noisy terminal observation $\delta = r_K + \zeta$, $\zeta \sim \mathcal{N}(0, \varepsilon_\gamma I)$,

$$\delta \mid r_i \sim \mathcal{N}(F_{i:K}r_i, (S_{i:K} + \varepsilon_\gamma)I). \quad (66)$$

The joint Gaussian moments are

$$\text{Cov}(r_i, \delta) = S_{0:i}F_{i:K}I, \quad \text{Var}(\delta) = (S_{0:K} + \varepsilon_\gamma)I. \quad (67)$$

Therefore,

$$\mathbb{E}[r_i \mid \delta] = \frac{S_{0:i}F_{i:K}}{S_{0:K} + \varepsilon_\gamma} \delta, \quad (68)$$

and

$$\text{Var}(r_i \mid \delta) = S_{0:i}I - \frac{S_{0:i}^2 F_{i:K}^2}{S_{0:K} + \varepsilon_\gamma} I \quad (69)$$

$$= \frac{S_{0:i}(S_{i:K} + \varepsilon_\gamma)}{S_{0:K} + \varepsilon_\gamma} I, \quad (70)$$

where the second equality uses $S_{0:K} = F_{i:K}^2 S_{0:i} + S_{i:K}$.

C One-Step Bridge Posterior Derivation

The posterior combines the passive transition

$$r_{i+1} \mid r_i \sim \mathcal{N}(A_i r_i, q_i^2 I) \quad (71)$$

with the terminal likelihood

$$\delta \mid r_{i+1} \sim \mathcal{N}(F_{i+1:K}r_{i+1}, (S_{i+1:K} + \varepsilon_\gamma)I). \quad (72)$$

For one coordinate, write $x = r_{i+1}$, $m = A_i r_i$, $a = F_{i+1:K}$, and $R = S_{i+1:K} + \varepsilon_\gamma$. The product density is proportional to

$$\exp \left(-\frac{(x - m)^2}{2q_i^2} - \frac{(\delta - ax)^2}{2R} \right). \quad (73)$$

The posterior variance is

$$C_i^\gamma = \frac{q_i^2 R}{R + q_i^2 a^2} = q_i^2 \frac{S_{i+1:K} + \varepsilon_\gamma}{S_{i:K} + \varepsilon_\gamma}, \quad (74)$$

where $S_{i:K} = q_i^2 F_{i+1:K}^2 + S_{i+1:K}$. The posterior mean is

$$\bar{r}_{i+1}^{\star, \gamma} = m + \frac{q_i^2 a}{R + q_i^2 a^2} (\delta - am) \quad (75)$$

$$= A_i r_i + \frac{q_i^2 F_{i+1:K}}{S_{i:K} + \varepsilon_\gamma} (\delta - F_{i:K} r_i). \quad (76)$$

This proves Proposition 3.

D SOC and Doob's h -Transform

Consider the controlled SDE

$$dr_t = \theta_t r_t dt + g_t u_t dt + g_t dW_t \quad (77)$$

and terminal cost $\frac{\gamma}{2}\|r_K - \delta\|^2$. The linearly-solvable SOC formulation with running cost $\frac{1}{2}\|u_t\|^2$ yields the desirability function

$$h_t^\gamma(r) = \mathbb{E}_R \left[\exp \left(-\frac{\gamma}{2}\|r_K - \delta\|^2 \right) \mid r_t = r \right]. \quad (78)$$

The optimal control is

$$u_t^{\star, \gamma} = g_t \nabla_r \log h_t^\gamma(r_t), \quad (79)$$

so the additional controlled drift is $g_t u_t^{\star, \gamma} = g_t^2 \nabla_r \log h_t^\gamma$. For the linear-Gaussian reference, the explicit form of h_t^γ derived in Section 3 gives Eq. (20). As $\gamma \rightarrow \infty$, this is the hard terminal conditioning limit, matching the Doob h -transform. For finite γ , it is the soft terminal controller used in the UniDB-style interpretation.

E Schedule and Progress Calibration

The bridge construction in Section 3 is valid for any positive sequence $\{\theta_i\}_{i=0}^{K-1}$. The schedule is therefore an inductive bias rather than a separate modeling component.

Legacy linear-beta schedule. The inherited diffusion-style schedule is

$$\theta_i = \frac{\beta_{\min}}{K} + \frac{\beta_{\max} - \beta_{\min}}{K^2}(i+1), \quad i = 0, \dots, K-1. \quad (80)$$

Prefix-calibrated schedule. Let $c_i \in [0, 1]$ be a desired hard-bridge marginal progress curve with $c_0 = 0$ and $c_K = 1$. Given a total rate $\Theta_{\text{tot}} > 0$, define

$$\Theta_i = \text{asinh}(c_i \sinh(\Theta_{\text{tot}})), \quad \theta_i = \Theta_{i+1} - \Theta_i. \quad (81)$$

For the hard bridge,

$$\beta_i = \frac{S_{0:i} F_{i:K}}{S_{0:K}} = \frac{\sinh(\Theta_i)}{\sinh(\Theta_K)}. \quad (82)$$

Therefore the above construction gives $\beta_i = c_i$, and hence

$$\mathbb{E}[s_{t+i} \mid s_t, s_{t+K}] = (1 - c_i)s_t + c_i s_{t+K}. \quad (83)$$

The default choice is $c_i = i/K$. Faster-prefix ablations may use $c_i = (i/K)^\alpha$ with $0 < \alpha < 1$.

F Prefix Schedule Derivation

For the hard bridge,

$$\beta_i = \frac{S_{0:i} F_{i:K}}{S_{0:K}}. \quad (84)$$

Let $\Theta_i = \Theta_{0:i}$ and $\Theta_K = \Theta_{\text{tot}}$. Since $S_{0:i} = \lambda^2(e^{2\Theta_i} - 1)$, $F_{i:K} = e^{\Theta_K - \Theta_i}$, and $S_{0:K} = \lambda^2(e^{2\Theta_K} - 1)$,

$$\beta_i = \frac{(e^{2\Theta_i} - 1)e^{\Theta_K - \Theta_i}}{e^{2\Theta_K} - 1} \quad (85)$$

$$= \frac{e^{\Theta_K}(e^{\Theta_i} - e^{-\Theta_i})}{e^{\Theta_K}(e^{\Theta_K} - e^{-\Theta_K})} \quad (86)$$

$$= \frac{\sinh(\Theta_i)}{\sinh(\Theta_K)}. \quad (87)$$

Thus setting

$$\Theta_i = \text{asinh}(c_i \sinh(\Theta_{\text{tot}})) \quad (88)$$

gives $\beta_i = c_i$. Since the bridge marginal mean is $(1 - \beta_i)s_0 + \beta_i s_K$, Proposition ?? follows.

G Critic and Actor Loss Details

Let Q_φ^H be a full chunk critic with horizon H , Q_φ^P a partial chunk critic with horizon h_a , and V_ξ a scalar goal-conditioned value function. A representative full chunk target is

$$y_t^H = \text{clip} \left(\sum_{k=0}^{H-1} \gamma^k r_{t+k} + \gamma^H V_\xi(s_{t+H}, s_g), 0, 1 \right), \quad (89)$$

with loss

$$\mathcal{L}_{Q^H} = \mathbb{E} \text{ BCE}(Q_\varphi^H(s_t, s_g, a_{t:t+H-1}), \text{sg}(y_t^H)). \quad (90)$$

The partial critic may be trained by optimistic distillation from the full critic:

$$\mathcal{L}_{Q^P} = \mathbb{E} \ell_{\kappa_d}(\text{sg}(Q_\varphi^H(s_t, s_g, a_{t:t+H-1})) - Q_\varphi^P(s_t, s_g, a_{t:t+h_a-1})), \quad (91)$$

where ℓ_{κ_d} is an expectile-type implicit maximization loss. The value function can be trained by a quantile or BCE-style implicit backup,

$$\mathcal{L}_V = \mathbb{E} \ell_{\kappa_b}(\text{sg}(Q_\varphi^P(s_t, s_g, a_{t:t+h_a-1})) - V_\xi(s_t, s_g)). \quad (92)$$

The SPI actor loss in Eq. (55) then optimizes the actor against Q^P while staying close to the Q-Boltzmann inverse-dynamics proposal distribution.

G.1 Derivation of the W_2 -SPI Actor Loss

We derive Eq. (55) from a Wasserstein proximal policy improvement problem. Fix a state-subgoal pair $(s_t, \hat{s}_K)$, and write

$$Q(a) := \bar{Q}_\varphi^{h_a}(s_t, \hat{s}_K, a).$$

Let the critic-induced energy of a policy distribution π be

$$\mathcal{E}(\pi) = -\mathbb{E}_{a \sim \pi}[Q(a)].$$

Given a reference proposal distribution π_{ref} , a single-step proximal policy improvement update under W_2 geometry is

$$\pi^+ \in \arg \min_{\pi} \left\{ -\mathbb{E}_{a \sim \pi}[Q(a)] + \frac{1}{2\tau} W_2^2(\pi, \pi_{\text{ref}}) \right\}.$$

In DOURI, the reference distribution is the Q-Boltzmann inverse-dynamics proposal

$$\pi_{\text{ref}} = \pi_{\text{invdyn}}^Q = \sum_{m=1}^M \rho_m \delta_{a_m},$$

where

$$a_m := \hat{a}_{0:h_a-1}^{(m)}, \quad \rho_m = \frac{\exp(\beta_{\text{sbi}} Q(a_m))}{\sum_{j=1}^M \exp(\beta_{\text{sbi}} Q(a_j))}.$$

The actor class is deterministic:

$$\pi_\phi = \delta_{\mu_\phi}.$$

Therefore the SPI objective restricted to this class is

$$\min_{\phi} \left\{ -Q(\mu_{\phi}) + \frac{1}{2\tau} W_2^2 \left(\delta_{\mu_{\phi}}, \sum_{m=1}^M \rho_m \delta_{a_m} \right) \right\}.$$

It remains to compute the Wasserstein term. By definition,

$$W_2^2(P, Q) = \inf_{\Gamma \in \Pi(P, Q)} \int \|x - y\|_2^2 d\Gamma(x, y),$$

where $\Pi(P, Q)$ is the set of couplings with marginals P and Q . For

$$P = \delta_{\mu}, \quad Q = \sum_{m=1}^M \rho_m \delta_{a_m},$$

the first marginal forces all mass to originate at the single point μ . The second marginal forces mass ρ_m to arrive at a_m . Hence the coupling is unique:

$$\Gamma^* = \sum_{m=1}^M \rho_m \delta_{(\mu, a_m)}.$$

Substituting this coupling gives

$$W_2^2 \left(\delta_{\mu}, \sum_{m=1}^M \rho_m \delta_{a_m} \right) = \sum_{m=1}^M \rho_m \|\mu - a_m\|_2^2.$$

Thus

$$\min_{\phi} \left\{ -Q(\mu_{\phi}) + \frac{1}{2\tau} \sum_{m=1}^M \rho_m \|\mu_{\phi} - a_m\|_2^2 \right\}.$$

Restoring the full conditioning variables and applying stop-gradient to the proposal weights and proposal actions yields Eq. (55).

H Notation Summary

Symbol	Definition	Role
s_0, s_K	current state, selected subgoal	bridge endpoints
r_i	$s_i - s_0$	displacement state
δ	$s_K - s_0$	terminal displacement
θ_i	positive integrated rate	schedule
g_i^2	$2\lambda^2 \theta_i$	continuous-time reference coefficient
q_i^2	$\lambda^2 (e^{2\theta_i} - 1)$	exact one-step variance
$\Theta_{i:j}$	$\sum_{\ell=i}^{j-1} \theta_{\ell}$	accumulated rate
$F_{i:j}$	$e^{\Theta_{i:j}}$	deterministic flow
$S_{i:j}$	$\lambda^2 (F_{i:j}^2 - 1)$	interval variance
β_i^{γ}	marginal coefficient	bridge progress
B_i^{γ}	marginal variance	bridge sampling scale
κ_i^{γ}	one-step terminal gain	posterior mean coefficient
C_i^{γ}	one-step posterior variance	residual/noise scale

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